Jimmy Pan

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Education

University of British Columbia **Expected May 2029**

BASc – Engineering Physics (GPA: 4.22)

Awards: Trek Excellence Scholarship (2025), Dean's Scholar (2025), Dean's List (2026)

Coursework: Instrument Design (Robotics), Computing Systems, Software Construction, Data Structures and Algorithms, Signals and Systems, Probability with Physical Applications, Applied Linear Algebra

Experience

UBC Rocket **Jul 2025 – Present**

Firmware Co-Lead Vancouver, BC

- Lead **STM32 flight-controller** firmware development, assign tasks, conduct detailed **PR reviews**, and coordinate architecture and integration with avionics hardware, software, and controls teams.
- Developed modular **I^2C interfaces** supporting blocking and interrupt-driven asynchronous transfers; separated device, bus, and STM32 HAL layers to improve portability and maintainability.
- Established firmware structure, API, and documentation standards; reviewed **SPI/UART drivers**, MCU configuration, **CMake** build changes, and **GitHub CI** workflows for code quality and consistency.
- Identified timeout, error-recovery, and peripheral-ownership risks in asynchronous I^2C code reviews; defined follow-up improvements applicable across communication drivers.
- Built a reusable **PWM servo-control driver** with STM32 HAL, hardware timers, and position-control APIs; refactored embedded utilities for reuse across control applications.
- Developed **C++/Qt Quick (QML)** Ground Control Station interfaces for telemetry visualization, system alerts, and logging; debugged serial communication and packaged the application for **Linux**.

Projects

Crashy – Dual-Core FPV Drone | *C, STM32, CMake, FreeRTOS, Kalman Filtering, PCB Design* **Oct 2026 – Present**

- Lead firmware architecture for a **dual-core STM32H755ZIT6** quadcopter, partitioning flight-critical Cortex-M7 processing from Cortex-M4 telemetry and supporting services.
- Built independent **Cortex-M7/M4 firmware** targets using STM32CubeMX, startup code, linker scripts, ARM GCC, and **CMake** with reproducible Debug/Release builds and per-core compile databases.
- Implemented **GitHub Actions CI** to cross-compile and verify both firmware images in Debug/Release, enforce clang-format, and support a documented Linux/VS Code development workflow.
- Planning **FreeRTOS** scheduling and inter-core communication, **Kalman-filter** sensor fusion, and custom **PCB design** for IMU/GPS acquisition and **DShot**-controlled ESC integration.

Crabby – Autonomous Mars Rover | *C/C++, ESP32-S3, UART/SPI/I^2C, PCB Design, Robotics* **May – Aug 2026**

- Led **embedded firmware architecture** for an autonomous competition robot with two **ESP32-S3** MCUs and a **Raspberry Pi**; placed **3rd overall** after demonstrating all five competition tasks.
- Developed **PWM/DIR/BRAKE motor-control**, encoder, servo, and tape-sensor drivers; integrated I^2C ToF sensors and stepper-motor control for coordinated movement.
- Designed a framed **UART protocol** linking both MCUs and the Raspberry Pi, with packet parsing, command handling, error recovery, and non-blocking autonomous execution.
- Integrated dual **PMW3610 optical sensors** using **SPI** for relative position and heading estimation; debugged sensor data alongside encoder-based motion control.
- Designed, assembled, and validated **four custom PCBs** for power distribution, motor control, H-bridge drivers, and tape sensing; tested hardware–software integration on the complete robot.

Closed-Loop RPM Motor Control System | *Digital Logic, Analog Circuits, PI Control, Circuit Debugging* **Nov – Dec 2025**

- Built and validated an **analog PI motor-speed controller** with a phototransistor, 8-bit counter, R-2R DAC, and BJT motor driver.
- Debugged latch/reset timing and frequency-to-voltage measurement circuits with **Digilent AD3**; characterized RPM regulation under changing setpoints and mechanical loads.

Skills

Programming & Software: C/C++, Python, Java, SystemVerilog, QML, MATLAB, Git/GitHub, CMake, Linux
Embedded & Control: STM32, ESP32-S3, UART, SPI, I^2C, PWM, Interrupts, DMA, Motor Control, Kalman Filtering
Digital & Hardware: FPGA Design, PCB Design, KiCad, Altium, Digilent AD3/WaveForms, Oscilloscopes, Logic Analyzers
Tools & Testing: VS Code, IntelliJ IDEA, Qt Creator, STM32CubeMX/IDE, PlatformIO, ESP-IDF, Quartus Prime, Questa, ARM GCC, GDB, OpenOCD, pyOCD, GitHub Actions, JUnit